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A text analytics model for agricultural knowledge discovery and sustainable food production: A case study from Oklahoma Panhandle

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ABSTRACT

With recent increases in the use of social media in agricultural communities, many farmers are showing more and more interest in participating in social media and sharing different aspects of their profession with peers and policymakers. However, researchers have not explored this valuable data source well enough to help improve agribusiness decision-making. This study aims to investigate the potential capability and richness of social media for agricultural knowledge discovery, which can help monitor, detect, and predict critical agricultural events and activities and develop more sustainable food production and agricultural economy. This research utilizes text-mining tools and techniques to collect, process, and mine unstructured textual data from Twitter. Then, it examines the agreement between the retrieved information from tweets and that from the current monitoring systems and common cultural practices. Our findings illustrate that social media data can effectively provide information regarding the commencement and duration of significant cultural activities This study also examines the classification performance of several popular sentiment analysis tools on the farmer's tweets and provides suggestions for future research on domain-specific sentiment lexicons for agricultural purposes.

1. Introduction

Previous studies have discussed concepts such as citizens as sensors [1] and social sensing [2,3] in the context of geographic and meteorological information, arguing that humans are intelligent sensors that constantly collect data about their surrounding environment. When human observations, perceptions, experiences, and feelings are shared through social media, a large amount of unstructured textual data is generated. Such data embed invaluable hidden knowledge which can be extremely beneficial in improving the economy and management of large and complex systems either at the individual or organizational levels.

Text mining as a knowledge discovery technique has been rapidly growing in the past two decades. It is defined as the process of discovering and extracting knowledge from unstructured or semi-structured textual datasets such as email, full-text documents, HTML files, and social media posts [4]. Frequency analysis of words, text classification, sentiment analysis, and topic extraction are the most popular text-mining tasks that have been utilized in the past in knowledge discovery in various domains such as information technology [5], natural resources [6], food science and nutrition [7], emergency response [8], marketing [9], tourism [10], finance [11], healthcare [12], and cybersecurity [13]. However, the application of text mining in the agricultural domain has remained somewhat limited. In a recent study, Drury and Roche [14] conducted a literature review and concluded that text mining is an underutilized approach in the agricultural domain.

[14] One of the earlier studies that implemented text mining techniques in agricultural applications was the work of Valsamidis et al. [15], which developed a sentiment analysis framework and applied it to an experimental agricultural blog. In another study, Jones-Garcia [16] employed speech recognition and machine translation to convert farmer's conversations to text and used text analysis techniques to ascertain the needs of farmers for agricultural extension services. Farmers' conversations were recorded through phone interviews. Sodoge et al. [17] proposed an approach for automated detection of drought impacts using newspaper articles that covered droughts in Germany from 2000 to 2021. Their method utilizes natural language processing (NLP) and machine learning to discern various socio-economic impacts, such as those on agriculture, forestry, livestock, and fires, along with their spatial and temporal dimensions. Mahon et al. [18] employed sentiment analysis and qualitative methods to investigate

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public sentiments regarding farming and the potential influence of visual components within agricultural images on these perceptions. They collected data through an online survey related to three images depicting modern UK dairy farms involving 521 individuals.

Social media text analytics is the process of extracting valuable insights and information from text data shared on social media platforms. This new field of study has successfully been applied to various domains such as healthcare [19], natural disasters [20], tourism [21], cryptocurrency [22], and sport [23]. Interestingly, most of the prior text analytics research in the agricultural domain has used social media text data. In a comprehensive study, Zipper [24] investigated the ability of Twitter in providing crop planting progress and discussed the challenges and opportunities of utilizing social media data in agricultural research, while acknowledging that only a limited number of studies have used Twitter data to monitor the spatiotemporal evolution of agricultural events. Shankar et al. [25] developed a framework for real-time monitoring of crop disease outbreaks based on Twitter data. Investigating spatiotemporal dynamics of other stressors such as drought based on social media data has been also carried out in several studies. Smith et al. [26] showed that tweets containing drought-related hashtags can serve as an effective tool in assessing drought impact and suggested that social media data should be considered in the future along with hydrometeorological indicators of drought. Pan et al. [27] gathered social media posts related to COVID-19 and its impact on agriculture from China's two largest media platforms, WeChat and Sina Weibo. They leveraged web crawler technology and text mining methods to investigate how COVID-19 has affected China's agricultural economy and the strategies employed to address these challenges. Riley and Robertson [28] studied how farmers utilize social media to capture and portray their daily work routines. By leveraging a dataset of tweets from the farming community, they demonstrated how social media can provide a more diverse, community-driven perspective of rural farming life. Shangguan et al. [29] gathered an extensive Twitter dataset related to climate change and performed topic modeling and natural language processing to unveil the relationship between the volume of climate change-related tweets and significant climate events. They also delved into the prevalent themes that individuals engage with regarding climate change and track the evolution of sentiment over time. Twitter data were also used by Zhang et al. [30] to study seven categories of drought impact (including agriculture). In a more recent study, Mukherjee et al. [31] employed machine learning methodologies on combined Twitter and satellite-derived drought data in Colorado. Three distinct machine-learning approaches were used to test the effectiveness of combining Twitter data with meteorological records to predict drought progression. The results indicated that integrating these data sources was a feasible approach, as the Twitter-inclusive model outperformed the model that lacked social media input. Also, Singh et al. [32] conducted a sentiment analysis on tweets posted by farmers during the farmers' protest in India. They investigated the sentiment polarity of farmers against farm regulations and policies as well as that of the public in different regions about the farmer's protest. With recent increases in the use and reach of social media in agricultural communities and with the heightened interest of many farmers in painting a more accurate and original picture of their profession by actively participating in social media (a.k.a. Agvocacy, [33]), a wealth of qualitative and quantitative data on different aspects of agricultural production is available in social media. However, the richness of such data and the utility of the existing text analysis tools (e.g., sentiment analysis lexicons) have not been well examined. Thus, this research aimed to answer the following questions: (1) Is social media text rich enough for agricultural knowledge discovery? (2) Are popular sentiment analysis lexicons capable enough for agricultural knowledge discovery? This research investigated the potential capability and richness of social media for agricultural knowledge discovery. We used text analysis to gain knowledge on several aspects of agricultural production, including planting, harvesting, irrigation management, irrigation systems, and drought. We considered these aspects because the potential gained knowledge can be of significant help in monitoring, early detection, and prediction of critical agricultural events and activities that help policymakers in developing more sustainable food production and agricultural economy.

We chose Oklahoma Panhandle for our case study because its economic and environmental sustainability is considerably impacted by the above-mentioned aspects. This is an agriculturally productive region under arid/semi-arid climate experiencing intensifying droughts and declining water resources. However, the same study can be conducted on other regions of interest.

This study analyzed tweets by farmers in Oklahoma Panhandle through both descriptive (temporal frequency and content analysis) and predictive (sentiment analyses) lenses that allow for a more accurate interpretation of observed patterns based on contextual information. The present study is different from the previous studies in this field in four ways. Firstly, we have restricted our social media data acquisition to only farmers, eliminating the issues caused by including tweets from news agencies and sales representatives. Secondly, we have not limited our keywords to hashtags, avoiding bias against the users who are not familiar with these hashtags [24]. Thirdly, while most previous papers have focused on only one aspect of agricultural production, we have investigated five (planting, harvesting, irrigating management, irrigation systems, and drought). Fourthly, we have evaluated the performance of several popular sentiment analysis tools on agricultural tweets.

The rest of the manuscript is organized as follows. In Sections 2.1 to 2.4, we describe the geographic area under study, our data sourcing and preparation steps. In Sections 2.5 and 2.6, we introduce our framework, including the descriptive and predictive analytics methods employed in this study. In Section 3, we apply our proposed framework to the prepared data and explain the results. In Section 4, we provide a discussion about the managerial implications of this study. Finally, in Section 5, we summarize our findings and conclude the manuscript by providing future research opportunities.

2. Materials and methods

In this section, we outline the key components of our research methodology. Fig. 1 schematically summarizes the implementation of the proposed framework. We begin with an explanation for the geographical area under study. Then, we describe our data acquisition process, detail data cleaning and preparation, and the techniques used for descriptive analysis and visualization. Lastly, we delve into sentiment analysis and try multiple lexicons on our data. These elements collectively underpin the foundation of our research.

2.1. Study area

This case study focused on the Oklahoma Panhandle, which is a major contributor to the state's agricultural production. The Oklahoma Panhandle has a semi-arid climate with normal annual reference evapotranspiration that is about 4.5 larger than the normal annual precipitation [34]. In addition, the region is prone to frequent droughts, which exacerbate the pressure on available water resources [35]. The Ogallala aquifer, which supplies 98% of the total water demand in the Oklahoma Panhandle has been declining continuously for the past several decades [36]. The largest user of water in the region is irrigated agriculture. Center pivot systems are by far the most common type of irrigation method, accounting for 95% of all irrigated areas in this region [37].

2.2. Data acquisition

To collect tweets, we identified 15 farmers on Twitter located within the Oklahoma Panhandle region by searching location and occupation keywords in Twitter user profiles. Since the Oklahoma Panhandle consists of five counties, our study relies on three observers per county.

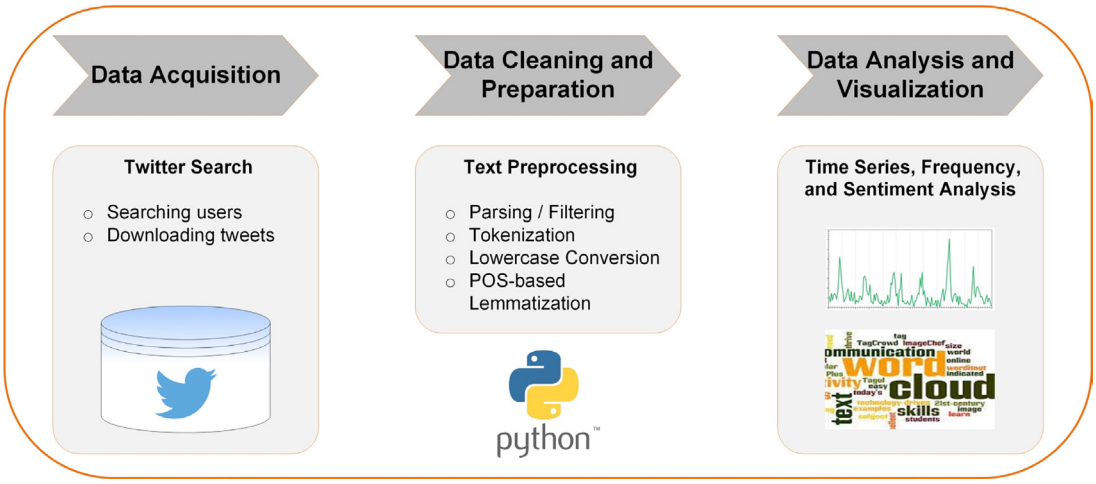

Fig. 1. Summary of the proposed framework.

Table 1
List of keywords used in searching farmers and tweets.

Keyword type	Keyword list
Location	Oklahoma Panhandle, OK Panhandle
Occupation	Farmer, Rancher, Cattleman, Producer, Agriculturalist
Domain specific	Plant, Harvest, Irrigat, Sprinkler, Pivot, Drought

This is considerably larger than the number of observers engaged in more traditional approaches of collecting agro-meteorological data. For example, the U.S. Drought Monitor relies on 450 experts from across the country [26], which translates to about to 0.15 observers per county. After identifying the users, their tweets were downloaded and retweets/duplicates were discarded, resulting in an initial dataset of over 130,000 tweets. The users had joined Twitter at different times, ranging from 2009 to 2014. We considered the tweets posted from January 1st, 2015 to December 30th, 2020, a period in which all users were active, and removed the remaining tweets, reducing the size of the initial dataset to 82,498 tweets. It is important to note that we systematically excluded any data pertaining to user identities to maintain the integrity of our analysis and prevent bias. Additionally, users were not made aware of the utilization of their tweets for this research, a precaution taken to ensure that there would be no influence on their future posts related to agriculture. Afterward, we searched for tweets containing five domain-specific keywords and stored them in separate databases for further investigation. Table 1 presents all the keywords used in collecting tweets.

2.3. Data cleaning and preparation

We employed a combination of Natural Language Processing (NLP) techniques to clean the collected tweets and convert them into a structured format. This process consisted of techniques such as transformation, tokenization, and normalization, all applied using the Natural Language Toolkit (NLTK) package in Python. The transformation step included the removal of all URLs, usernames, stop words (frequent words providing no information, such as “a”, “the”, “so”, “and”, etc.), and non-alphabetic characters like emojis, punctuations, and numbers. Then, through tokenization, we split the raw tweets into minimal units of text, named tokens. Finally, we normalized tokens through lowercase conversion and lemmatization to obtain a unique representation of them and reduce redundancy in the data. Lemmatization converts all the inflected forms of individual text tokens into their standard or root form. For example, harvest, harvesting, and harvested are different forms of harvest. Moreover, in our lemmatization process, we applied part-of-speech (POS) tagging, which is labeling tokens according to their part of speech, such as noun, verb, adjective, and so on, within a text. So, tokens are lemmatized based on their POS tags.

2.4. Domain specific searches

Domain-specific searches were conducted to identify tweets in five different categories: planting, harvesting, irrigation management, irrigation systems, and drought.

2.4.1. Planting, harvesting, irrigation management

Tweets related to three practices of planting, harvesting, and irrigating were obtained using the three search phrases “plant”, “harvest”, and “irrigat”. Temporal dynamics of relative frequencies of tweets related to each of these practices were investigated and compared against common practices in the Oklahoma Panhandle. In addition, recorded data on planting and harvest dates in each of the six years of study were obtained from two Oklahoma State University’s research farms in the region. These farms represent the general cultural practices in the Oklahoma Panhandle as the research projects are carried out in collaboration with local farmers and the results are intended to benefit these stakeholders.

2.4.2. Irrigation systems

All tweets related to center pivot irrigation systems were extracted, using the keywords “sprinkler” and “pivot” along with “WetWitch”, a term commonly used by local farmers when referring to center pivots. This database was then searched to find tweets that were related to four major types of system components or issues that require regular attention. These four types (and the keywords used to find their tweets) were: 1. water supply (pump, engine, motor), 2. water application (nozzle), 3. tower drive (gearbox), and 4. tower movement (track, rut, stuck).

2.4.3. Drought

Tweets about drought were identified using “drought” and “dryness” as keywords. Temporal dynamics of drought tweets were analyzed along with variations in the Drought Severity and Coverage Index (DSCI), downloaded for the study area (Oklahoma Panhandle) and weekly periods from the U.S. Drought Monitor website. The DSCI translates categorical drought data to a single value ranging from zero under no drought to 500 when the entire area of interest is under exceptional drought [38]. Drought categories of the U.S. Drought Monitor are based on objective and subjective data that 450 observers provide on the condition and impacts of drought. Drought tweets identified in the present study reflect the experience of farmers in the Oklahoma Panhandle with drought conditions and impacts. Hence, we were interested in examining the correlation between DSCI and drought tweets. The DSCI has been used before by Smith et al. [26] to predict the number of tweets containing #drought.